

tf.compat.v1.uniform_unit_scaling_initializer

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https://www.tensorflow.org/api_docs/python/tf/compat/v1/uniform_unit_scaling_initializer

Initializer that generates tensors without scaling variance.

Function Signatures

```
tf.compat.v1.uniform_unit_scaling_initializer( factor=1.0,  
seed=None, dtype=tf.dtypes.float32 )
```

Code Examples

```
tf.compat.v1.uniform_unit_scaling_initializer(  
    factor=1.0,  
    seed=None,  
    dtype=tf.dtypes.float32  
)
```

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    factor=1.0,  
    seed=None,  
    dtype=tf.dtypes.float32  
)
```

```
tf.dtypes.float32
```

```
[-sqrt(3) / sqrt(dim), sqrt(3) / sqrt(dim)]
```

```
[-sqrt(3) / sqrt(dim), sqrt(3) / sqrt(dim)]
```

```
@classmethod  
from_config(  
    config  
)
```

```
@classmethod
```

```
from_config(  
    config  
)
```

Documentation

Initializer that generates tensors without scaling variance.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.initializers.uniform_unit_scaling`

When initializing a deep network, it is in principle advantageous to keep the scale of the input variance constant, so it does not explode or diminish by reaching the final layer. If the input is x and the operation $x * W$, and we want to initialize W uniformly at random, we need to pick W from

to keep the scale intact, where $\text{dim} = W.\text{shape}[0]$ (the size of the input). A similar calculation for convolutional networks gives an analogous result with dim equal to the product of the first 3 dimensions. When nonlinearities are present, we need to multiply this by a constant factor. See (Sussillo et al., 2014) for deeper motivation, experiments and the calculation of constants. In section 2.3 there, the constants were numerically computed: for a linear layer it's 1.0, relu: ~ 1.43 , tanh: ~ 1.15 .

References

Methods

from_config

[View source](#)

Instantiates an initializer from a configuration dictionary.

Example:

```
## get_config
```

[View source](#)

Returns the configuration of the initializer as a JSON-serializable dict.

__call__

[View source](#)

Returns a tensor object initialized as specified by the initializer.